

Why are you making this chart?

Before a single colour is chosen or a single bar is drawn, every good data visualisation answers one question: what is it for? This module is the 14th key-knowledge point of the Data Analysis chapter — the purposes of data visualisations.

What this key knowledge covers

This summary distils everything you need for **KK14 — Why are you making this chart?** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Purpose is decided before you design

A common beginner mistake is to make a chart and *then* ask what it's for. The study design flips that. Purpose is part of the **solution requirements** — it's settled in the **Analysis** stage of the problem-solving methodology, long before you open the software. Purpose then drives every later design choice: chart type, colours, fonts, how much text, how the data is framed.

Think of a film director. They don't film random footage and then decide whether it's a comedy or a documentary. They decide the *genre* first, and every shot serves it. Your visualisation's purpose is its genre. Pick it first; let it command the rest.

Purpose answers two questions

Who is the audience (their knowledge, their stake)? and **what do you want them to walk away with** — understanding, a feeling, the facts, or a decision?

One visualisation, one dominant purpose

A visualisation can do a little of several things, but in an exam you name the **primary** purpose and justify it. "It informs *and* persuades" without choosing a dominant one usually loses marks.

The four purposes

Memorise these four — and crucially, the *difference* between them. A quick hook: "**Every Eager Intern Persuades**" → **E**ducate, **E**ntertain, **I**nform, **P**ersuade.

Change minds or behaviour

Educate = "now I understand it." **Entertain** = "that was fun." **Inform** = "now I know the facts." **Persuade** = "now I want to act." If you can finish the sentence the audience would say,

you've found the purpose.

Lab — the Purpose Prism

Here is the central idea in one picture. The GreenLoop dataset (left) enters a prism and splits into four outputs — one per purpose. **Choose a purpose** below and watch the *same data* get reshaped for a different job. Notice: the numbers never change; the **design** does.

Lab — how a fact becomes a weapon

The line between **inform** and **persuade** is the most-tested idea in this dot point — and it often comes down to one design choice: **where the axis starts**. Below is GreenLoop's recycling rate for four weeks: 62%, 64%, 66%, 69%. A real, modest rise. Toggle the y-axis baseline and watch the same numbers go from "honest" to "dramatic".

A truncated (cut-off) axis isn't always "wrong" — but it **shifts the purpose toward persuade**, because it's chosen to create an impression rather than to report neutrally. If a question asks you to evaluate whether a visualisation *informs*, a misleading axis is your evidence that it leans toward **persuade** instead.

The subtle pair — educate vs inform

Both deal in facts and aim to be accurate, so students mix them up. The split is about **what the audience does with it**.

Educate → teaches

Builds **understanding of how/why**. The audience comes away *able to do or explain* something. Tends to be sequenced, annotated, with the "mechanism" shown.

Tell-tale: arrows, steps, labels explaining a process; "here's how it works".

Inform → reports

States **the current facts / state of affairs**. The audience comes away *knowing what is*, not necessarily how it works. Tends to be a snapshot or trend.

Tell-tale: dashboards, status charts, "here's where things stand right now".

Educate: a flow showing *how* a plastic bottle is sorted, washed, shredded and re-pelletised.

Inform: a chart showing *how many* bottles were collected each week. One teaches the process; the other reports the count.

The dangerous pair — spotting persuade

Persuade is the purpose with an **agenda**. The data may be accurate, but it's *framed* to push you toward a viewpoint or an action. Learn the design "tells" — these are exactly what examiners reward you for naming.

① A point of view

It's clearly "for" or "against" something. Inform stays neutral; persuade takes a side.

② A call to action

"Switch now", "Vote yes", "Bring a reusable". An explicit ask to the audience is a dead giveaway.

③ **Emotive language & colour**

Alarming reds, big bold words, shock numbers ("9,000 bottles!") chosen to make you *feel*.

④ **Selective / framed data**

Truncated axes, cherry-picked time ranges, "1 in 3" framing — the data that supports the case is foregrounded.

A persuasive visualisation can be completely truthful. Persuasion is about **intent and framing**, not falsehood. When framing crosses into distorting the facts, it becomes **misleading** — but the underlying purpose is still to persuade.

Name that purpose

Eight real-world style visualisations. Read each, then tap the purpose you think dominates. You'll get instant feedback and the examiner's reasoning. Every correct answer adds to your score dock.

Match the design choice to the purpose

Purpose drives design. Tap a **design choice** on the left, then the **purpose** it best serves on the right. Correct pairs lock in.

Educate

"Now I understand how/why it works." Teaches a concept or skill with steps and annotations.

Entertain

"That was fun!" Playful, personalised, emotion-first — precision is secondary.

Inform

"Now I know the current facts." Neutral, accurate, lets the audience judge.

Persuade

"Now I want to act." A viewpoint + call to action, framed to move you.

Inform vs Persuade

A call to action or selective framing → persuade. Neutral, balanced, honest axes → inform.

Where is purpose decided?

Analysis. Purpose is part of the solution requirements and drives every later design choice.

Six exam traps

The mistakes that quietly cost marks. Read them now so you don't make them under pressure.

P·E·L writing trainer

"Identify and justify the purpose" questions reward a tight structure. Use **P-E-L: Point** (name the purpose) → **Evidence** (a design feature you can see) → **Link** (how that feature serves that purpose for that audience). Draft each part; the meter fills as your answer gains the keywords examiners look for.

GreenLoop releases a bright red poster reading "9,000 bottles wasted this term. Bring a reusable — start Monday." with a giant arrow to a pledge form. **Identify the purpose of this visualisation and justify your answer with reference to its design. (3 marks)**

VCE-style exam questions

Five questions, **17 marks** total. Try each yourself first. Then reveal a weak answer, a strong answer and the marking guide — and self-mark honestly. Your total tracks in the dock.

- 1 mark for naming two correct purposes.
- 2 marks for naming all four: educate, entertain, inform, persuade.
- No marks for descriptions like 'to look nice' instead of a named purpose.
- 1 mark: correctly identifies the purpose as inform.
- 1 mark: first valid design feature (e.g. neutral/honest axes, plain labels).
- 1 mark: second valid feature OR explaining the absence of persuasion (no call to action/emotive framing).
- Identifying inform with no design evidence caps the answer at 1 mark.
- 1 mark: A identified as educate.
- 1 mark: design feature of A linked to educating (steps/labels teaching how).
- 1 mark: B identified as persuade.
- 1 mark: design feature of B linked to persuading (call to action / emotive shock figure).
- Top answers note the same data serves different purposes.
- 1 mark: identifies the axis as truncated / not starting at zero.
- 1 mark: explains the visual effect (small change exaggerated).
- 1 mark: links this to a shift in purpose toward persuade (away from inform).
- 1 mark: states the risk — misleading the audience / distorted decisions / lost credibility.
- 1 mark: judgement that the statement is incorrect / too absolute.
- 1 mark: explains persuade is a valid purpose based on framing/intent, not falsehood.
- 1 mark: gives a supporting example (truthful but persuasive).
- 1 mark: acknowledges persuasion CAN become misleading, distinguishing framing from distortion.

Key terms

Term	Meaning
Data visualisation	A graphical representation of data designed to reduce the effort needed to interpret it — charts, graphs, infographics and dashboards.
Infographic	A designed combination of text and graphics that presents information or a story; commonly aims to inform, educate or persuade.
Dashboard	A visualisation that brings several charts/metrics together to show a current state at a glance — typically an <i>inform</i> purpose.

Term	Meaning
Call to action	An explicit prompt telling the audience to do something; a hallmark of a <i>persuade</i> purpose.
Truncated axis	An axis that doesn't start at zero, exaggerating differences. Shifts a visualisation from neutral (inform) toward (persuade), and can mislead.

Common traps & misconceptions

Exam trap

TRAP 01 Confusing **inform** and **persuade** because both "show data". **Fix:** ask "*is it neutral, or does it want me to act?*" A call to action or framing = persuade.

Exam trap

TRAP 02 Saying a visualisation "does all four purposes" to play it safe. **Fix:** name the *dominant* purpose and justify it. Hedging reads as not knowing.

Exam trap

TRAP 03 Treating **educate** and **inform** as identical because both are factual. **Fix:** educate teaches *how/why* (understanding); inform reports *what* (current facts).

Exam trap

TRAP 04 Thinking **persuade** means the data is fake or wrong. **Fix:** persuasion is about *framing & intent*. Truthful data can still be persuasive.

Exam trap

TRAP 05 Identifying a purpose but giving *no evidence* from the visualisation. **Fix:** always quote a design feature — colour, axis, call to action, annotations — as your proof.

Exam trap

TRAP 06 Forgetting **entertain** exists, so labelling a playful "Wrapped"-style card as inform. **Fix:** personalised, fun, emotion-first design with loose precision = entertain.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. State

The four purposes of a data visualisation.

✗ A weak answer

To make data look nice and to show numbers. To inform people and to make graphs colourful.
Examiner: Only one of four purposes is named ('inform'); the rest are vague descriptions, not purposes.

✓ A strong answer

The four purposes of a data visualisation are to **educate, entertain, inform** and **persuade** an audience.

How the marks are awarded

- 1 mark for naming two correct purposes.
- 2 marks for naming all four: educate, entertain, inform, persuade.
- No marks for descriptions like 'to look nice' instead of a named purpose.

Q2. Identify • 3 marks

A health service publishes a clean dashboard showing current emergency-department wait times by hospital, with honest axes and plain labels. Identify the purpose and justify your answer with reference to two design features. () Tip: identify first, then quote two visible features as evidence.

✗ A weak answer

The purpose is to inform because it shows data. It has numbers and colours so people can read it. Examiner: Correct purpose, but 'shows data' and 'has numbers and colours' aren't specific design features tied to informing.

✓ A strong answer

The purpose is **to inform**. Evidence: (1) the **honest, neutral axes and plain labels** present the facts without exaggeration, and (2) it shows the **current state** (live wait times) for the audience to read and judge for themselves. There is no call to action or emotive framing, so it reports rather than persuades.

How the marks are awarded

- 1 mark: correctly identifies the purpose as inform.
- 1 mark: first valid design feature (e.g. neutral/honest axes, plain labels).
- 1 mark: second valid feature OR explaining the absence of persuasion (no call to action/emotive framing).
- Identifying inform with no design evidence caps the answer at 1 mark.

Q3. Compare · 4 marks

Two visualisations use the same recycling dataset. Visualisation A is a labelled flow diagram showing each step of how plastic is recycled. Visualisation B is a bold poster reading "9,000 bottles wasted — switch to reusables today". Compare the purposes of A and B, referring to design features of each. () This is a compare question — make sure you address both, and contrast them.

✗ A weak answer

A and B are both about recycling. A teaches you and B tells you to recycle. They both use data about bottles. Examiner: Names a rough idea for each but uses no purpose vocabulary and almost no design features.

✓ A strong answer

Visualisation A's purpose is to educate. Its **labelled, sequenced flow diagram** teaches the audience *how* recycling works step by step, building understanding. **Visualisation B's purpose is to persuade.** Its **bold emotive styling, the shock figure "9,000 bottles wasted" and the explicit call to action ("switch to reusables today")** are designed to change the audience's behaviour. Although both use the same data, A aims for understanding while B aims for action — purpose, not data, is what differs.

How the marks are awarded

- 1 mark: A identified as educate.
- 1 mark: design feature of A linked to educating (steps/labels teaching how).
- 1 mark: B identified as persuade.
- 1 mark: design feature of B linked to persuading (call to action / emotive shock figure).
- Top answers note the same data serves different purposes.

Q4. Explain · 4 marks

A student team makes a chart of recycling rates (62%, 64%, 66%, 69%) but starts the y-axis at 60% so week 4 looks far taller than week 1. Explain how this design choice affects the purpose of the visualisation, and the risk it creates. () Link the truncated axis to a shift in purpose, then to a risk.

✗ A weak answer

Starting the axis at 60% makes the bars bigger. This is bad because it's not at zero. It changes the chart. Examiner: Notices the effect but never names a purpose shift or the real risk (misleading the audience).

✓ A strong answer

Starting the axis at 60% **truncates** it, so a small 7-point rise is **visually exaggerated**. This shifts the visualisation away from a neutral **inform** purpose toward **persuade** — it now creates the impression "we're improving fast" rather than reporting the modest real change. The **risk** is that it **misleads** the audience: they may over-estimate the improvement and make decisions based on a distorted picture, which also damages the credibility of the team's data.

How the marks are awarded

- 1 mark: identifies the axis as truncated / not starting at zero.
- 1 mark: explains the visual effect (small change exaggerated).
- 1 mark: links this to a shift in purpose toward persuade (away from inform).
- 1 mark: states the risk — misleading the audience / distorted decisions / lost credibility.